

The Length-Zero (“Zero-Tail”) Bookkeeping in the Blocked Canonical-Form Decomposition

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This note explains why carrying the blocked canonical-form decomposition as an *all-length* matrix-product-vector identity is the wrong formal shape, and records the clean replacement together with the one place where the formal development still keeps the awkward form.¹

1 The assertion

Let A be a translation-invariant matrix-product tensor of bond dimension D , with one-site matrices $A^i \in M_D(\mathbb{C})$. For a length N and a word $\sigma = (i_1, \dots, i_N)$ write the matrix-product-vector coefficient

$$V^{(N)}(A)_\sigma = \text{tr}(A^{i_1} \cdots A^{i_N}). \quad (1)$$

The blocked canonical-form reduction of [CPGSV16, Section 2.3], following [PGVWC07, Theorem 3], states that after blocking by some period $p \geq 1$ the tensor is, up to a gauge, the direct sum of an all-zero block and a weighted family of normal blocks,

$$A^{[p]} \cong \mathbf{0}_z \oplus \bigoplus_k \mu_k B_k, \quad (2)$$

where $\mathbf{0}_z$ is the zero tensor on a bond space of dimension z , the B_k are normal (primitive, trace-preserving, irreducible) blocks of bond dimension D_k , the weights satisfy $\mu_k \neq 0$, and $z + \sum_k D_k = D$ is the bond dimension of $A^{[p]}$.

2 The point at issue

The direct sum (2) gives, at every length N and word σ ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\mathbf{0}_z)_\sigma + V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma. \quad (3)$$

¹Context: PRs PR #2153, PR #2156, and PR #2167 removed roughly a thousand lines of “zero-tail” plumbing built on the all-length form discussed here. This note records why that form is a non-source-faithful overcomplication and what remains.

The zero block contributes nothing except at the empty word: for $N \geq 1$ a product of A^i -blocks that passes through the zero summand vanishes, so

$$V^{(N)}(\mathbf{0}_z)_\sigma = 0 \quad (N \geq 1), \quad (4)$$

whereas at $N = 0$ the only coefficient is the trace of the identity on the bond space,

$$V^{(0)}(C)_\emptyset = \text{tr}(\mathbf{1}_{D_C}) = D_C. \quad (5)$$

Hence the *entire* content of the length-zero instance of (3) is the dimension count

$$z + \sum_k D_k = D, \quad (6)$$

which is already part of (2). Every length $N \geq 1$ carries the genuine mathematical content: the weighted normal-block comparison $V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma$.

Why the all-length form is the wrong shape. Stating the reduction as the all-length identity (3) attaches the trivial dimension fact (6) to the substantive positive-length comparison and forces both to travel together through blocking, reindexing, and sector grouping. In the formal development this produced a family of transport lemmas whose only purpose was to move the empty-word instance of (3) across each such step.² This is also not source-faithful: the article introduces the zero block once, to account for the bond dimension, and then discards it, comparing only the nonzero normal blocks in the thermodynamic limit. The zero summand $\mathbf{0}_z$ is an artifact of keeping a common bond space in the formal statement; it carries no state and no normal-block structure.

3 The clean statement

Separate the two facts. The positive-length comparison

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma \quad (N \geq 1) \quad (7)$$

carries all of the mathematics, and the length-zero coefficient is recovered *once*, where an all-length statement is genuinely needed, from the single dimension identity (6): if two tensors agree at all positive lengths and have equal bond dimension, they agree at length zero as well, because the empty-word coefficient is the bond dimension.³ This is the source-faithful shape: the nonzero normal blocks are compared at positive length, and the zero block appears only as the scalar z in (6).

²These were the `zeroTail*` lemmas in `TNLean/MPS/CanonicalForm/SectorComparison/`; they are removed in PRs PR #2156 and PR #2167 once the comparison is stated at positive length.

³Formally, positive-length equality is `SameMPVPos` and the one-step upgrade is `SameMPVPos.toSameMPV'of'bondDim'eq` in `TNLean/MPS/Defs`. The canonical-form predicate now stores `SameMPVPos` together with a single bond-dimension field rather than the all-length decomposition; see PR #2153.

4 Formal status

exists `tp_primitive_blockDecomp` after blocking, blueprint `thm:tp_primitive_blockdecomp` in `blueprint/src/chapter/ch11b_after_blocking.tex`, now uses the source-faithful shape: its MPV clause is the positive-length comparison in (7), and the empty-word contribution is the separate identity $z + \sum_k D_k = D$. The former support lemma `zeroTailToTensorFromBlocks_blockPower` has therefore been removed.

5 Conclusion

The blocked canonical form (2) is correct, and the dimension count (6) is part of it. The former formal defect was one of shape: stating the reduction as the all-length matrix-product-vector identity (3) fused a trivial empty-word dimension count with the substantive positive-length comparison. The faithful statement is the positive-length comparison (7) with the length-zero coefficient restored once from (6). The development now uses this shape in the after-blocking primitive decomposition and its downstream consumers.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.
- [PGVWC07] David Pérez-García, Frank Verstraete, Michael M. Wolf, and J. Ignacio Cirac. Matrix product state representations. *Quantum Information and Computation*, 7(5–6):401–430, 2007.